# Dinh Duy Kha

Postdoctoral Researcher @ University of British Columbia, Canada
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## Introduction

I am a postdoctoral researcher at the University of British Columbia with a Ph.D. from Sungkyunkwan University. I build practical and principled solutions at the intersection of operating systems and security. My current research focuses on compartmentalization, side-channel mitigations, confidential computing, and policy compliance. My work has been recognized with the ACM CCS Distinguished Paper Award (2023) and published at top security venues including IEEE S&P and USENIX Security.

## Education

Sungkyunkwan University — Suwon, South Korea
*Ph.D. in Computer Science* — 2019 - 2026

- Thesis title: "Retrofitting Operating System Security for Modern Hardware Protections"
- Synopsis: Argues that protections established by hardware and OS must be coherent to close protection gaps exploited by modern attackers. Presents two hardware-OS coherent designs: Capacity [3], which enforces capability-based intra-process memory isolation via ARM Pointer Authentication and Memory Tagging, and IncognitOS [2], which redesigns OS scheduling and memory management to suppress side-channel leakage in confidential VMs.

Ho Chi Minh City University of Technology — Ho Chi Minh City, Vietnam
*B.S. in Computer Science* — 2014 - 2019

## Work Experience

Systopia Lab, University of British Columbia — Vancouver, BC, Canada
*Postdoctoral Researcher* — 2026 - present

System Security Lab, Sungkyunkwan University — Suwon, South Korea
*Graduate Student Researcher* — 2019 - 2026

VNG Corporation — Ho Chi Minh City, Vietnam
*Software Engineering Intern – TalkTV Backend* — Jun 2018 - Jun 2019

## Awards

Noteworthy Reviewer Recognition, USENIX Security Artifact Evaluation — 2025
Korean Government BK21 Research Scholarship — 2024
Distinguished Paper Award, ACM CCS — 2023

## Projects

Cryptographic capabilities for program compartmentalization [3]

- Designed an in-process capability-based compartmentalization framework leveraging ARM Pointer Authentication and Memory Tagging; evaluated its security properties.
- Developed LLVM compiler instrumentation passes to enforce memory access security policies.
- Developed a kernel module to support isolation of file-related objects using the capability-based framework.

Full-system memory obfuscation for confidential virtual machines [2]

- Designed a unikernel-based execution environment providing full-system side-channel obfuscation for confidential VMs (CVMs).

- Implemented adaptive memory rerandomization in the paging subsystem via direct MMU access, adjusting rerandomization frequency based on observed VMExit rates.
- Evaluated resilience against CVM-targeted side-channel attacks and demonstrated practical performance for real-world workloads.

LLM-assisted C to Rust translation

- Investigated developers practices for translating C into Rust from real-world repositories
- Constructed LLM-assisted C-to-Rust translation pipeline based on C2Rust.

Simulation framework for PIM-based confidential computing [1]

- Used gem5 to develop a full-system simulation of emerging PIM hardware.
- Implemented and evaluated security features to enable PIM as an accelerator for confidential computing.

## Teaching

*Graduate Teaching Assistant*, Sungkyunkwan University

- ESW4010: Special Topics on System Security (Fall 2021, Spring 2022, Fall 2022)
  - Developed an automated framework to deploy CTF challenges to docker containers.
  - Adapted the CTFd framework to make it more suitable for the classroom environment.
  - The framework was used to teach subsequent courses at `https://ctf.skku.edu/`.
- SWE2001: System Programming (Fall 2021)

## Academic Service

- **Artifact Evaluation Committee:** USENIX Security '25, ACM CCS '25
- **Poster Program Committee:** ACM CCS '25

## Open Source Contributions

- Unikraft – implemented AMD SEV (Secure Encrypted Virtualization) support and submitted various bug fixes to the core kernel

## Skills

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|---|---|
| **Languages** | Assembly, C, C++, Rust, Python, Bash |
| **Systems** | KVM, QEMU, Linux kernel, device drivers, LLVM/Clang, gem5, Intel SGX, AMD SEV |

## Publications


[1] Woo, B., **Duy, K. D.**, Han, Y., Kang, B. B., Lee, H., "Pim-oram: Towards oblivious ram primitives in commodity processing-in-memory," in *2025 IEEE Annual Computer Security Applications Conference (ACSAC)*, Honolulu, Hawaii, USA: ACM, Dec. 2025. [Online]. Available: `https://ieeexplore.ieee.org/document/11391727`.

[2] **Duy, K. D.**, Kim, J., Lim, H., Lee, H., " IncognitOS: A Practical Unikernel Design for Full-System Obfuscation in Confidential Virtual Machines," in *2025 IEEE Symposium on Security and Privacy (SP)*, Los Alamitos, CA, USA: IEEE Computer Society, May 2025, pp. 4192–4209. doi: `10.1109/SP61157.2025.00222`. [Online]. Available: `https://doi.ieeecomputersociety.org/10.1109/SP61157.2025.00222`.

[3] Cho, K., Kim, J., **Duy, K. D.**, Lim, H., Lee, H., "RustSan: Retrofitting AddressSanitizer for efficient sanitization of rust," in *33rd USENIX Security Symposium (USENIX Security 24)*, Philadelphia, PA: USENIX Association, Aug. 2024, pp. 3729–3746, ISBN: 978-1-939133-44-1. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/cho-kyuwon.

[4] **Duy, K. D.**, Cho, K., Noh, T., Lee, H., "Capacity: Cryptographically-Enforced In-Process Capabilities for Modern ARM Architectures," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '23, Copenhagen, Denmark: Association for Computing Machinery, 2023, pp. 874–888, ISBN: 9798400700507. DOI: 10.1145/3576915.3623079. [Online]. Available: https://doi.org/10.1145/3576915.3623079.

[5] **Duy, K. D.**, Lee, H., "Se-pim: In-memory acceleration of data-intensive confidential computing," *IEEE Transactions on Cloud Computing*, vol. 11, no. 3, pp. 2473–2490, 2023. DOI: 10.1109/TCC.2022.3207145.

[6] **Duy, K. D.**, Noh, T., Huh, S., Lee, H., "Confidential machine learning computation in untrusted environments: A systems security perspective," *IEEE Access*, vol. 9, pp. 168 656–168 677, 2021. DOI: 10.1109/ACCESS.2021.3136889.